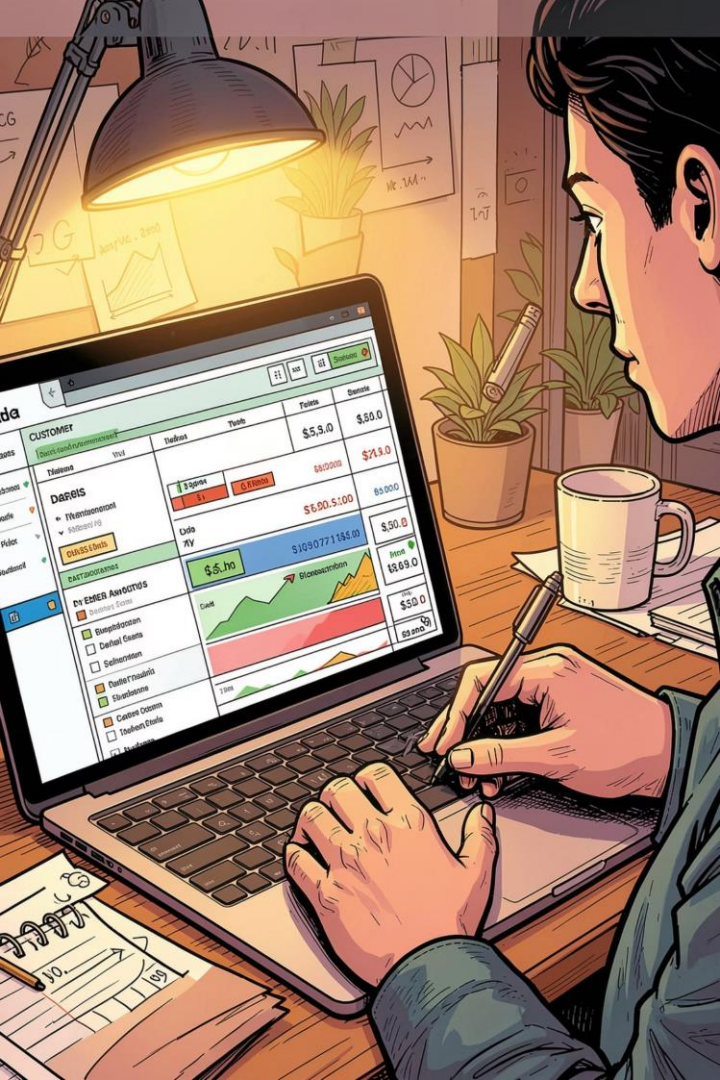




Customer Shopping Behavior Analysis

3,900 purchases — 18 features — goal: uncover spend patterns, segments, product & subscription insights

Ciatatels			Sumbcls	Slampel
 Radrans	\$00	\$4593	12,383%	2,8,3%
 Resaris	4,203	\$,365	34%	90%
 Firio	39%	38%	20%	\$3,38%
 Dacarmait	\$02	\$115		4,38%
 Cadads	355	2,3845		3,56%
 Batioenben	320	37%	18%	10,35%
 Bashaine	18%	\$56	40%	\$,58%
 Alldetr	\$45	\$39	41%	\$2,3,35%
Satmon	338	\$55		\$,460%
Peinials	\$98	\$9%		
 Ceabviadcato	\$,79%	3,6%		1,79%
Dsaned	3,038	30%	4%	3,44%
 Fino	1,746	\$93	—	\$,28%
 Dapohlaciadenty	\$,087	3%	9%	\$,78%
 Nirdohtsach	16	\$,38%	18%	\$3,35%
 Bacl	1.150	\$517	20%	\$,56%
 Eajnadrcation	62	\$311	40%	\$2,76%

Dataset Summary

Rows & Columns

Rows: 3,900 ·
Columns: 18

Key Features

Demographics,
Purchase details,
Behavior, Shipping

Missing Data

37 missing values
in Review Rating

Exploratory Data Analysis (Python)

01

Load & Inspect

pandas, df.info(), describe()

02

Missing Values

Imputed Review Rating by category
median

03

Standardize

Renamed columns to snake_case

04

Feature Eng.

age_group, purchase_frequency_days

05

DB Integration

Loaded cleaned DF to PostgreSQL

SQL Analysis — Key Metrics



Revenue by Gender

Female: 75,191 · Male: 157,890

Avg Purchase

\$59.76 overall



Discounts & High-Spenders



High-Spending Discount Users

839 rows identified — customers using discounts but spending above average



Discount-Dependent Products

Top: Hat (50%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), Pants (47.37%)

Top Products & Categories



Top Rated Items

Gloves 3.86 · Sandals 3.84 · Boots 3.82 · Hat 3.80 · Skirt 3.78



Top by Orders

Accessories: Jewelry, Sunglasses, Belt ·
Clothing: Blouse, Pants, Shirt · Footwear:
Sandals, Shoes, Sneakers

Shipping & Spend



Shipping Comparison

Standard avg: \$58.46 · Express avg: \$60.48

Subscribers vs Non

Subscribers: 1,053 customers · Avg spend \$59.49 · Revenue \$62,645

Non-subscribers: 2,847 customers · Avg spend \$59.87 · Revenue \$170,436

Customer Segmentation & Repeat Buyers

Loyal

3,116 customers

Returning

701 customers

New

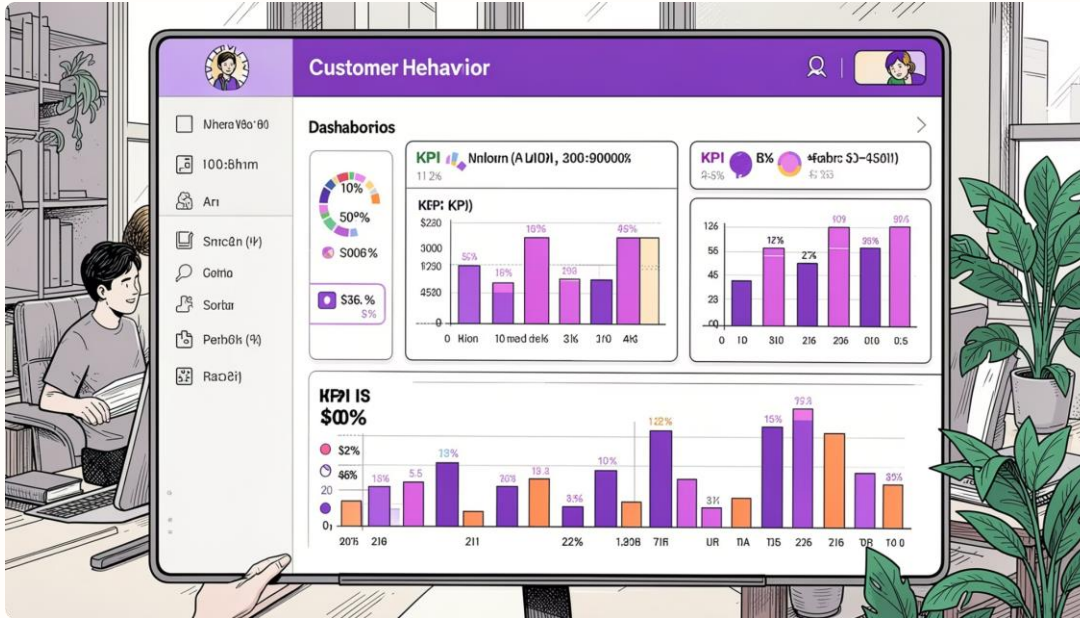
83 customers

Repeat buyers (>5 purchases): 958 — check subscription propensity

SEGMENTOMR ICGROUPS



Power BI Dashboard



KPIs

3.9K customers · \$59.76 avg purchase ·
3.75 avg rating

Visuals

Subscription donut (27% yes) · Revenue
& Sales by Category · Revenue & Sales
by Age Group

Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to increase loyalty

Review Discounts

Balance sales lift with margin control

Product Positioning

Highlight top-rated & best-sellers in campaigns

Targeted Marketing

Focus on high-revenue age groups & express-shipping users